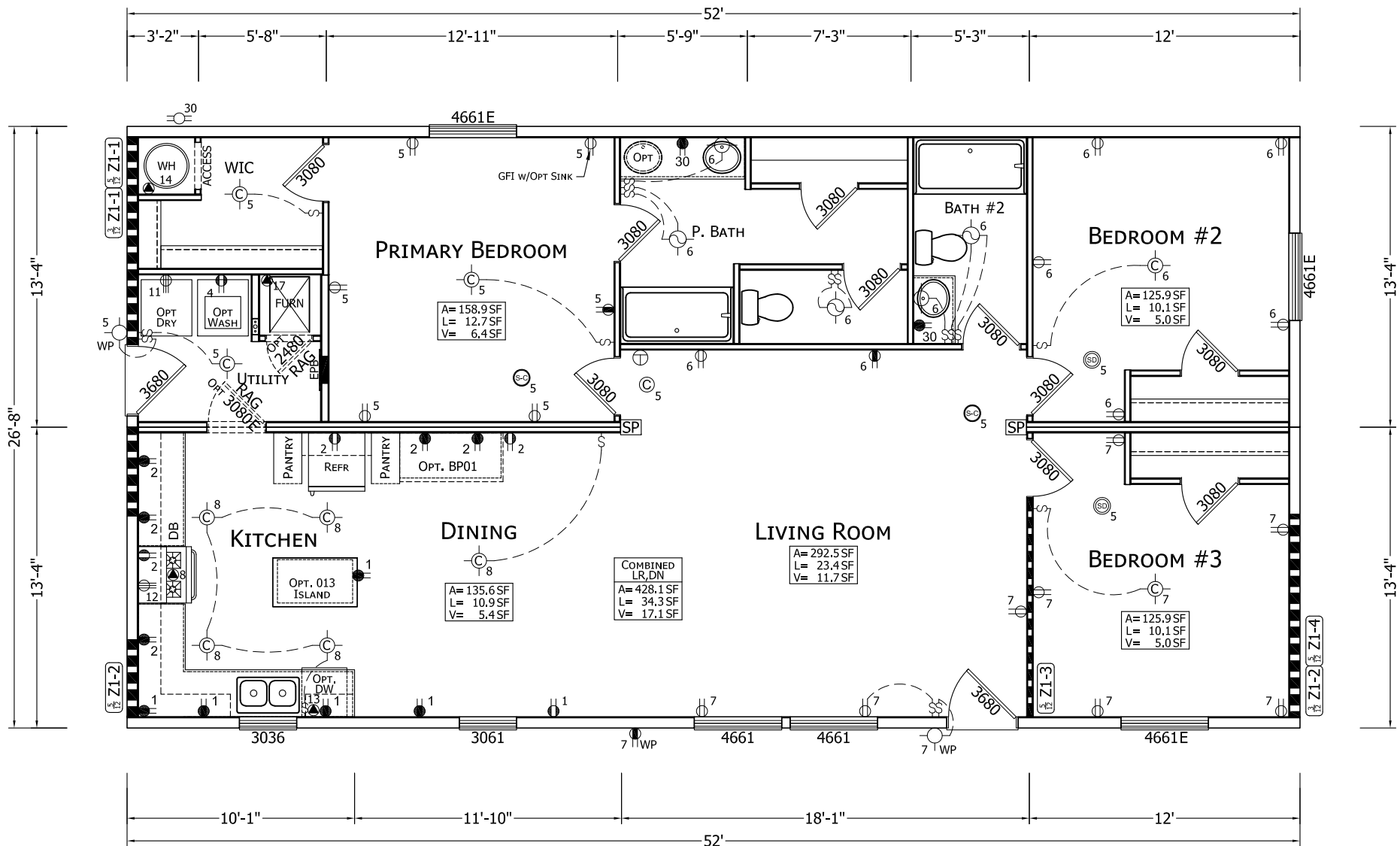




DESIGN CRITERIA
ZONES: WIND 1, ROOF MIDDLE, TEMP III
TOTAL ROUGH WINDOW OPENINGS: 120.2 SF

CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



1 2025 CODE UPDATE

MODIFICATIONS

3/12/2025 JR

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT

DRAWN BY: JON RILEY

DATE: 10-21-24

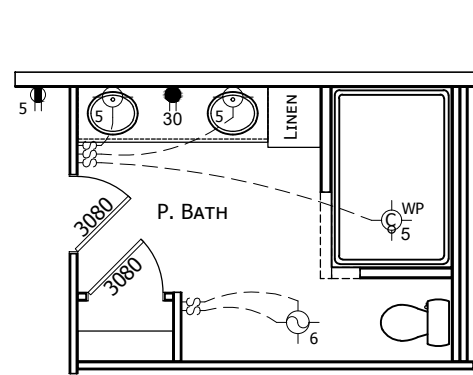
SCALE: 5/32" = 1'-0"

TITLE:
FLOOR PLAN
ELECTRICAL PLAN

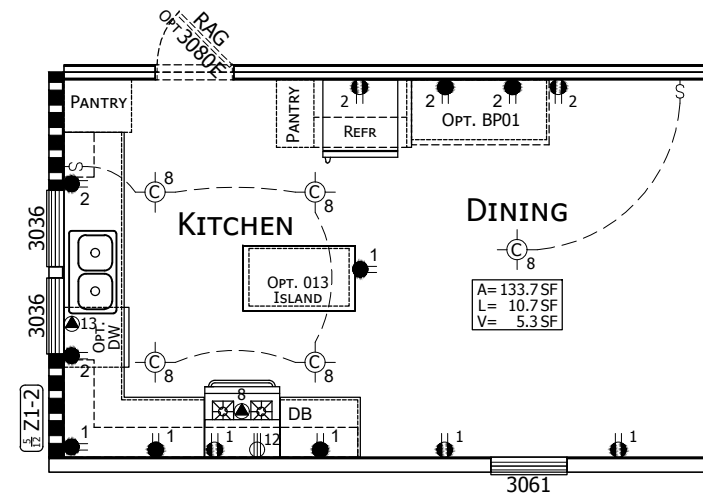
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SHEET:
AE-101

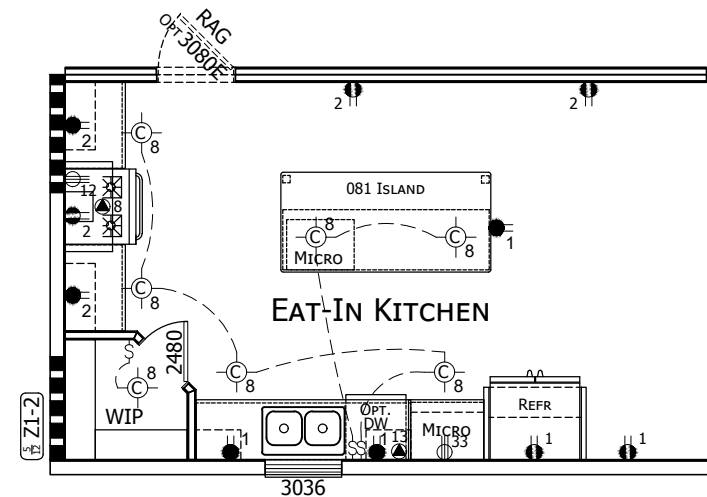
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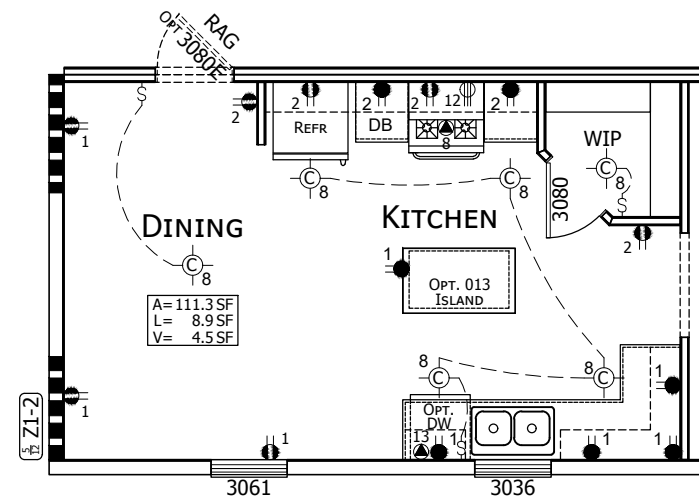
OPTIONAL SERENITY BATH



OPTIONAL KITCHEN #1



OPTIONAL SUMMIT KITCHEN
AVAILABLE W/ OPT. STONE COUNTERTOPS



OPTIONAL KITCHEN #2

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

1 2025 CODE UPDATE

3/12/2025 JR

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT

DRAWN BY: JON RILEY
DATE: 10-21-24
SCALE: 5/32" = 1'-0"

TITLE:
FLOOR PLAN
ELECTRICAL PLAN

FILENAME: 112APB-2852H32170

SHEET:
AE-102

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Overall Check = PASS

CHB Shear Wall - Layout / Design

Model#: 112APB-2852H32170

Wind Zone = I

Wind Load = 15 psf

Module Width = 160 in

Number of Boxes = 2

Wall Height = 96 in

Roof Pitch = 4.36 :12

Angle = 20.0 °

Roof Overhang = 8 in

Floor Joist Spacing = 16 in o.c.

Rim Joist Plies = 1

Frame Spacing = 99.5 in

Beam Depth = 8 in

Open Porch = 0.00 ft.

Home Length = 52.00 ft.

Diaphragm Construction:

Diaphragm Type = Wood Roof

Fasteners 15GAx1.5

Minimum Width = in

Max plf = 211.6 plf

Max. End Wall Reaction = 5,643 lbs

Max. Interior Reaction = 11,285 lbs

Max. Span = 112.00 ft.

Floor Construction:

Floor Joist Size = 2x6

Floor Joist Species = SPF

Floor Joist Grade = #3

Decking Glued to Joists = Yes

Wall Construction:

Bottom Plate Thickness = 2x

Top Plate Thickness = 2x

Wall to Floor Joist = #10 x 3.5"

Wall to Ceiling Framing = #10 x 3.5"

Wall to Wall Fastener = #10 x 3.50"

Add 2 Shearwalls = No

Ground Tie Chart

Peir Height (in.)	Near Beam Angle (Deg.)	Far Beam Angle (Deg.)	Ground Tie Spacing (ft)	Max. Spacing (ft.)
12	37.30		12.37	12'-4"
20	46.85		10.64	10'-7"
28	53.90		9.25	9'-2"
36	59.18		7.97	7'-11"
44	63.22	22.47	14.37	14'-4"
52	66.37	25.51	14.04	14'-0"
60	68.89	28.40	13.68	13'-8"
68	70.95	31.15	13.31	13'-3"

* For monoslope homes only

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Left End Wall Shear Wall

Dist. To Next SW = 52.00 ft.

Manual Input = --

Load to Shearwall = 1560.0 lbs

Shear Wall 2

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 1560.0 lbs

Shear Wall 3

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 0.0 lbs

Shear Wall 4

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 0.0 lbs

Shear Wall 5

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 0.0 lbs

Shear Wall 6

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 0.0 lbs

Panel Layout

From Module Edge = 6.00 in.

Panel Length = 108.00 in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = 173.3 plf

Shearwall Panel = Gypsum

Gyp. Brand = U.S.G.

Fasteners = Yes

Adhesive = PVA

One Side or Two = 1

Max. Shearwall plf = 247 plf

Panel Layout

From Module Edge = 6.00 in.

Panel Length = 108.00 in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = 173.3 plf

Shearwall Panel = Gypsum

Gyp. Brand = U.S.G.

Fasteners = Yes

Adhesive = PVA

One Side or Two = 1

Max. Shearwall plf = 247 plf

Panel Layout

From Module Edge = in.

Panel Length = in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = N/A plf

Shearwall Panel =

Gyp. Brand =

Fasteners =

Adhesive =

One Side or Two =

Max. Shearwall plf = plf

Panel Layout

From Module Edge = in.

Panel Length = in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = N/A plf

Shearwall Panel =

Gyp. Brand =

Fasteners =

Adhesive =

One Side or Two =

Max. Shearwall plf = plf

Panel Layout

From Module Edge = in.

Panel Length = in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = N/A plf

Shearwall Panel =

Gyp. Brand =

Fasteners =

Adhesive =

One Side or Two =

Max. Shearwall plf = plf

Panel Layout

From Module Edge = in.

Panel Length = in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = N/A plf

Shearwall Panel =

Gyp. Brand =

Fasteners =

Adhesive =

One Side or Two =

Max. Shearwall plf = plf

*Construct per Note 6 on SW01.35.XX

Zone - ID Z1-1

Joist 2

Wall Conn. 8

SC O.C. 16 @ 7"

SF O.C. 18 @ 6"

Beam Lag 2

*May be 1 joist per Note 9.C. on SW01.00.04

Result = PASS

*Construct per Note 6 on SW01.35.XX

Zone - ID Z1-2

Joist 2

Wall Conn. 8

SC O.C. 16 @ 7"

SF O.C. 18 @ 6"

Beam Lag 2

*May be 1 joist per Note 9.C on SW01.00.04

Result = PASS

*Construct per SW01.25.01

Zone - ID

Joist

Wall Conn.

SC O.C.

SF O.C.

Beam Lag

Result =

*Construct per SW01.25.01

Zone - ID

Joist

Wall Conn.

SC O.C.

SF O.C.

Beam Lag

Result =

*Construct per SW01.25.01

Zone - ID

Joist

Wall Conn.

SC O.C.

SF O.C.

Beam Lag

Result =

*Construct per SW01.25.01

Zone - ID

Joist

Wall Conn.

SC O.C.

SF O.C.

Beam Lag

Result =

Overall Check = PASSCHB Shear Wall - Layout / DesignModel#:112APB-2852H32170

Wind Zone = IWind Load = 15 psfModule Width = 160 inNumber of Boxes = 2Wall Height = 96 inRoof Pitch = 5:12Angle = 22.6 °Roof Overhang = 8 inFloor Joist Spacing = 16 in o.c.Rim Joist Plies = 1Frame Spacing = 99.5 inBeam Depth = 8 inOpen Porch = 0.00 ftHome Length = 52.00 ft

Diaphragm Construction:Diaphragm Type = Gypsum CeilingFoam Brand = F-2100Minimum Width = 160 inMax plf = 219 plfMax. End Wall Reaction = 5,840 lbsMax. Interior Reaction = 11,680 lbsMax. Span = 48.00 ft

Floor Construction:Floor Joist Size = 2x6Floor Joist Species = SPFFloor Joist Grade = #3Decking Glued to Joists = Yes

Wall Construction:Bottom Plate Thickness = 2xTop Plate Thickness = 2xWall to Floor Joist = #10 x 3.5"Wall to Ceiling Framing = #10 x 3.5"Wall to Wall Fastener = #10 x 3.50"Add 2 Shearwalls = No

Ground Tie Chart

Peir Height (in.)	Near Beam Angle (Deg.)	Far Beam Angle (Deg.)	Ground Tie Spacing (ft)	Max. Spacing (ft.)
12	37.30		12.37	12'-4"
20	46.85		10.64	10'-7"
28	53.90		9.25	9'-2"
36	59.18		7.97	7'-11"
44	63.22	22.47	14.37	14'-4"
52	66.37	25.51	14.04	14'-0"
60	68.89	28.40	13.68	13'-8"
68	70.95	31.15	13.31	13'-3"

Vertical Trib = 9.56 ftKing Post Height *= in.Porch End Wall = --

Vertical Trib = 9.56 ftKing Post Height *= in.Porch End Wall = --

Vertical Trib = 9.56 ftKing Post Height *= in.Porch End Wall = --

Vertical Trib = 9.56 ftKing Post Height *= in.Porch End Wall = --

Vertical Trib = 9.56 ftKing Post Height *= in.Porch End Wall = --

Vertical Trib = 9.56 ftKing Post Height *= in.Porch End Wall = --

Left End Wall Shear WallDist. To Next SW = 0.00 ftManual Input = TRUELoad to Shearwall = N/A lbsReaction Override = 1589.3 lbs

Shear Wall 2Dist. To Next SW = 40.00 ftManual Input = TRUELoad to Shearwall = N/A lbsReaction Override = 1278.7 lbs

Shear Wall 3Dist. To Next SW = 12.00 ftManual Input = --Load to Shearwall = 3726.7 lbs

Shear Wall 4Dist. To Next SW = ftManual Input = --Load to Shearwall = 860.0 lbs

Shear Wall 5Dist. To Next SW = ftManual Input = --Load to Shearwall = 0.0 lbs

Shear Wall 6Dist. To Next SW = ftManual Input = --Load to Shearwall = 0.0 lbs

Panel LayoutFrom Module Edge = 6.00 inPanel Length = 110.00 inFrom Module Edge = inPanel Length = in

Panel LayoutFrom Module Edge = 6.00 inPanel Length = 41.25 inFrom Module Edge = 112.75 inPanel Length = 47.25 in

Panel LayoutFrom Module Edge = 6.00 inPanel Length = 116.50 inFrom Module Edge = inPanel Length = in

Panel LayoutFrom Module Edge = 6.00 inPanel Length = 96.00 inFrom Module Edge = inPanel Length = in

Panel LayoutFrom Module Edge = inPanel Length = inFrom Module Edge = inPanel Length = in

Panel LayoutFrom Module Edge = inPanel Length = inFrom Module Edge = inPanel Length = in

Shearwall plf = 173.4 plfShearwall Panel = GypsumGyp. Brand = U.S.G.Fasteners = YesAdhesive = PVAOne Side or Two = 1Max. Shearwall plf = 247 plf

Shearwall plf = 173.4 plfShearwall Panel = GypsumGyp. Brand = U.S.G.Fasteners = YesAdhesive = PVAOne Side or Two = 1Max. Shearwall plf = 247 plf

Shearwall plf = 383.9 plfShearwall Panel = GypsumGyp. Brand = U.S.G.Fasteners = YesAdhesive = PVAOne Side or Two = 2Max. Shearwall plf = 432 plf

Shearwall plf = 107.5 plfShearwall Panel = GypsumGyp. Brand = U.S.G.Fasteners = YesAdhesive = PVAOne Side or Two = 1Max. Shearwall plf = 247 plf

Shearwall plf = N/A plfShearwall Panel = GypsumGyp. Brand = U.S.G.Fasteners = YesAdhesive = PVAOne Side or Two = 1Max. Shearwall plf = 247 plf

Shearwall plf = N/A plfShearwall Panel = GypsumGyp. Brand = U.S.G.Fasteners = YesAdhesive = PVAOne Side or Two = 1Max. Shearwall plf = 247 plf

*Construct per Note 6 on SW01.35.XX

*Construct per Note 6 on SW01.35.XX

*Construct per Note 6 on SW01.35.XX

*Construct per Note 6 on SW01.35.XX

*Construct per SW01.25.01

*Construct per SW01.25.01

Zone - ID	Z1-1
Joist	2
Wall Conn.	8
SC O.C.	19 @ 6"
SF O.C.	19 @ 6"
Beam Lag	2

Zone - ID	Z1-2
Joist	2
Wall Conn.	8
SC O.C.	15 @ 6"
SF O.C.	17 @ 5.5"
Beam Lag	2

Zone - ID	Z1-3
Joist	3
Wall Conn.	8
SC O.C.	39 @ 3"
SF O.C.	47 @ 2.5"
Beam Lag	5

Zone - ID	Z1-4
Joist	2
Wall Conn.	8
SC O.C.	10 @ 10"
SF O.C.	12 @ 8"
Beam Lag	1

Zone - ID	
Joist	
Wall Conn.	
SC O.C.	
SF O.C.	
Beam Lag	

Zone - ID	
Joist	
Wall Conn.	
SC O.C.	
SF O.C.	
Beam Lag	

*May be 1 joist per Note 9.C. on SW01.00.04

*May be 1 joist per Note 9.C on SW01.00.04

Result = PASS

Result = PASS

Result = PASS

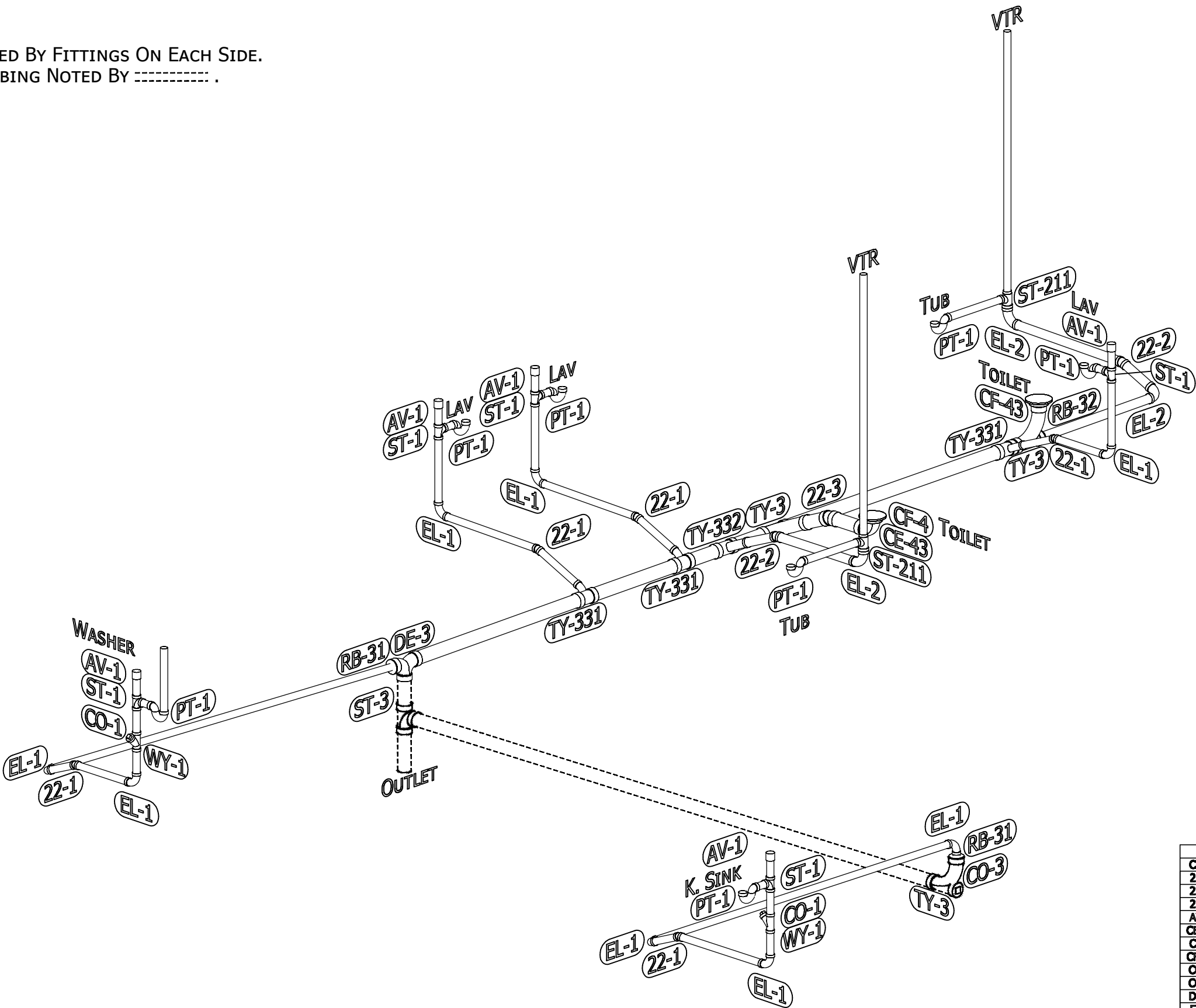
Result = PASS

Result =

Result =

NOTES:

- 1. PIPE SIZES DETERMINED BY FITTINGS ON EACH SIDE.
- 2. SITE INSTALLED PLUMBING NOTED BY -----.



FITTING SCHEDULE		FITTING SCHEDULE	
CALL	NAME	CALL	NAME
22-1	1 1/2" 22 1/2° ELL	EL-2	2" LT 90° ELL
22-2	2" 22 1/2° ELL	PT-1	1 1/2" P-TRAP
22-3	3" 22 1/2° ELL	RB-31	3" x 1 1/2" BUSHING
AV-1	1 1/2" AUTO VENT	RB-32	3" x 2" BUSHING
CE-43	4" x 3" CLOSET ELL	ST-1	1 1/2" TEE
CF-4	4" CLOSET FLANGE	ST-211	2" x 1 1/2" x 1 1/2" TEE
CF-43	4" x 3" CLOSET FLANGE	ST-3	3" TEE
CO-1	1 1/2" CLEANOUT	TY-3	3" LTYY
CO-3	3" CLEANOUT	TY-331	3" x 3" x 1 1/2" LTYY
DE-3	3" DOUBLE ELL	TY-332	3" x 3" x 2" LTYY
EL-1	1 1/2" LT 90° ELL	WY-1	1 1/2" WYE

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200

PFS^{#23}

Conforms to
HUD MHCSS

12/1/25

MODIFICATIONS

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT

DRAWN BY: JON RILEY

DATE: 10-21-24

SCALE: N.T.S.

TITLE:
DRAIN LINE
PLAN

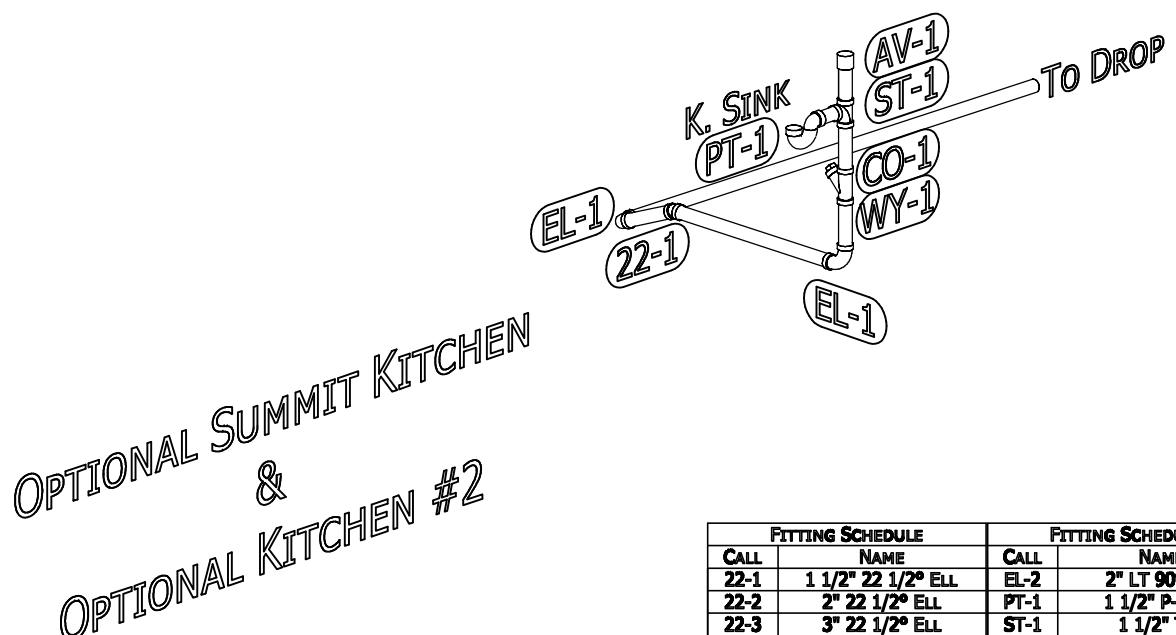
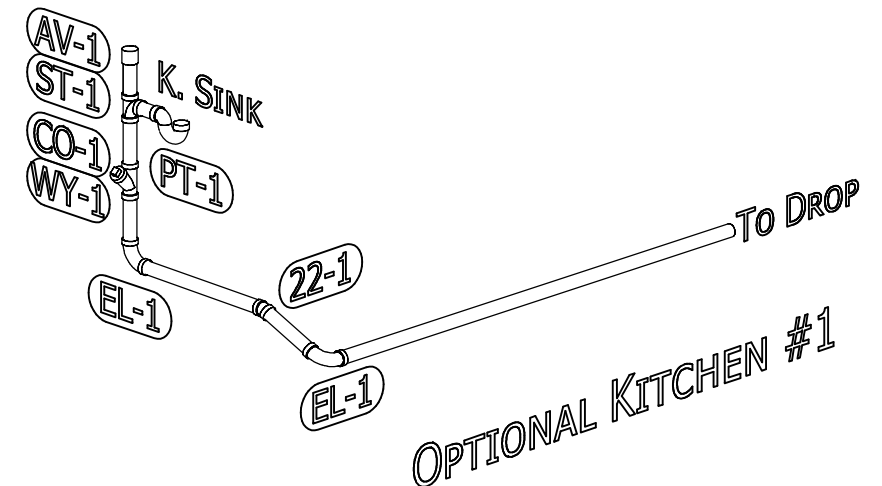
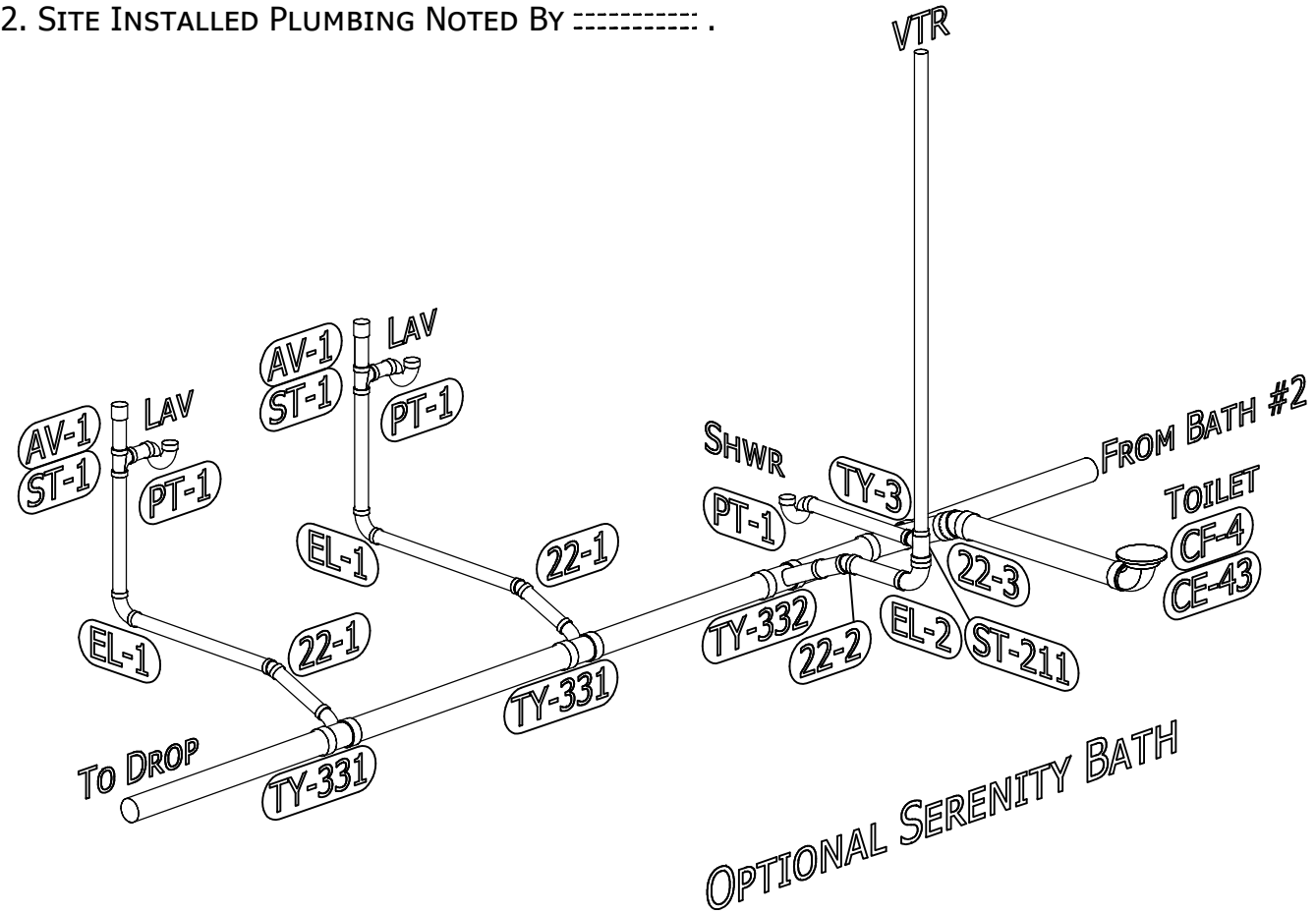
FILENAME: 112APB-2852H32170

SHEET:
D-101

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NOTES:

- 1. PIPE SIZES DETERMINED BY FITTINGS ON EACH SIDE.
- 2. SITE INSTALLED PLUMBING NOTED BY ----- .



FITTING SCHEDULE		FITTING SCHEDULE	
CALL	NAME	CALL	NAME
22-1	1 1/2" 22 1/2° ELL	EL-2	2" LT 90° ELL
22-2	2" 22 1/2° ELL	PT-1	1 1/2" P-TRAP
22-3	3" 22 1/2° ELL	ST-1	1 1/2" TEE
AV-1	1 1/2" AUTO VENT	ST-211	2" x 1 1/2" x 1 1/2" TEE
CE-43	4" x 3" CLOSET ELL	TY-3	3" LTYY
CF-4	4" CLOSET FLANGE	TY-331	3" x 3" x 1 1/2" LTYY
CO-1	1 1/2" CLEANOUT	TY-332	3" x 3" x 2" LTYY
EL-1	1 1/2" LT 90° ELL	WY-1	1 1/2" WYE

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200

PFS#23

Conforms to
HUD MHCSS
12/1/25

MODIFICATIONS

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT

DRAWN BY: JON RILEY
DATE: 10-21-24
SCALE: N.T.S.

TITLE:
DRAIN LINE
PLAN

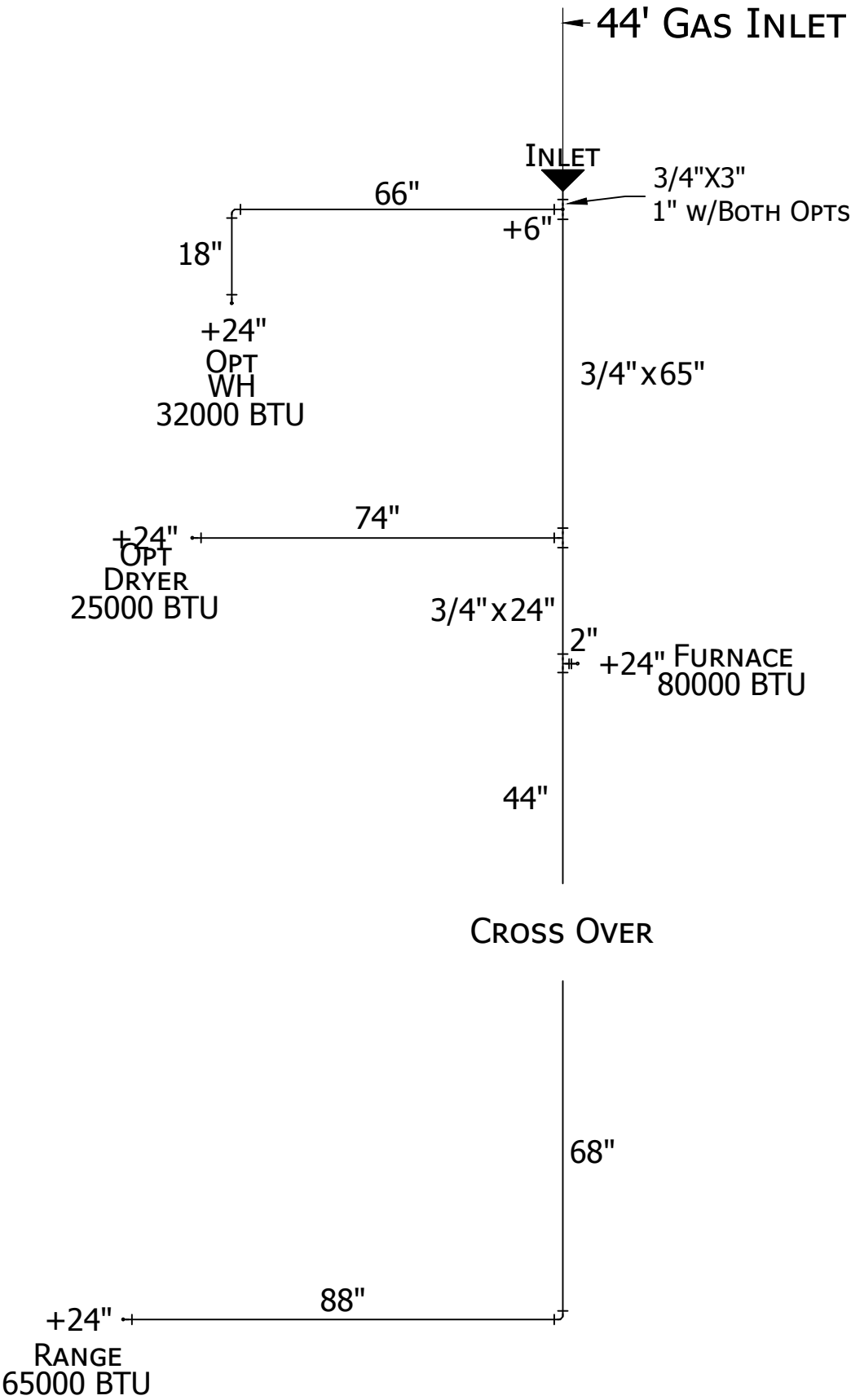
FILENAME: 112APB-2852H32170

SHEET:
D-102

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NOTES:

- 1. ALL PIPE SIZES 1/2" UNLESS OTHERWISE SPECIFIED
- 2. TOTAL BTU = 145,000 (202,000 W/ OPTIONS)
- 3. COLUMN LENGTH = 30'-0"



CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT
DRAWN BY: JON RILEY
DATE: 10-21-24
SCALE: N.T.S.

TITLE:
GAS LINE
PLAN

FILENAME: 112APB-2852H32170

SHEET:
G-101

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- 3. COLUMN LENGTH = 30'-0"

CROSS OVER

11'-8"

18"
2"± +6"
+24"
RANGE
65000 BTU

OPTIONAL KITCHEN #1

CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT
DRAWN BY: JON RILEY
DATE: 10-21-24
SCALE: N.T.S.

TITLE:
GAS LINE
PLAN

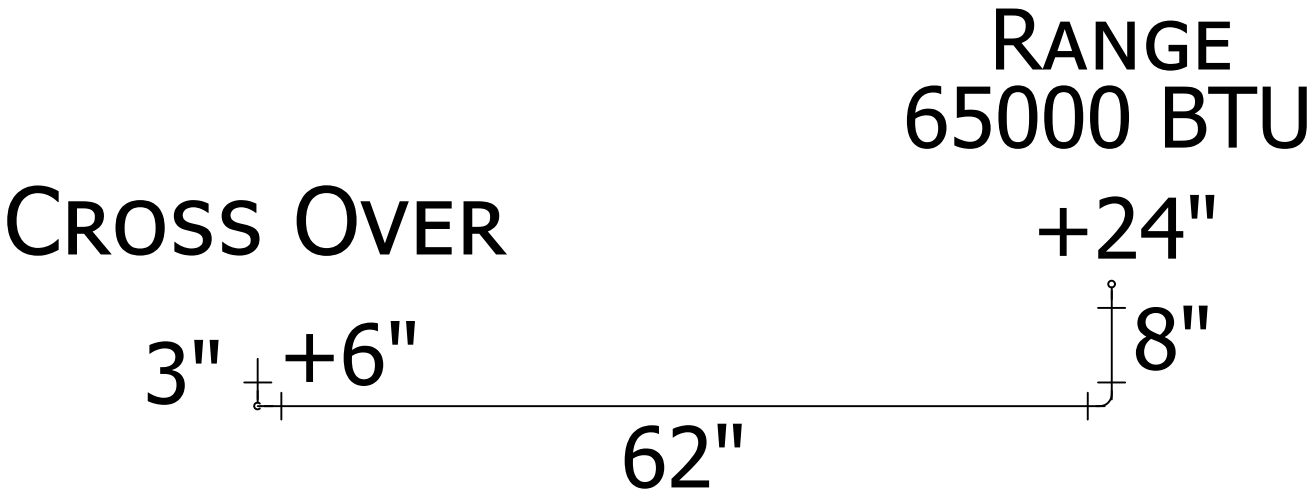
FILENAME: 112APB-2852H32170

SHEET:
G-102

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NOTES:

- 1. ALL PIPE SIZES 1/2" UNLESS OTHERWISE SPECIFIED
- 2. TOTAL BTU = 145,000 (202,000 W/ OPTIONS)
- 3. COLUMN LENGTH = 30'-0"



OPTIONAL KITCHEN #2

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT

DRAWN BY: JON RILEY
DATE: 10-21-24
SCALE: N.T.S.

TITLE:
GAS LINE
PLAN

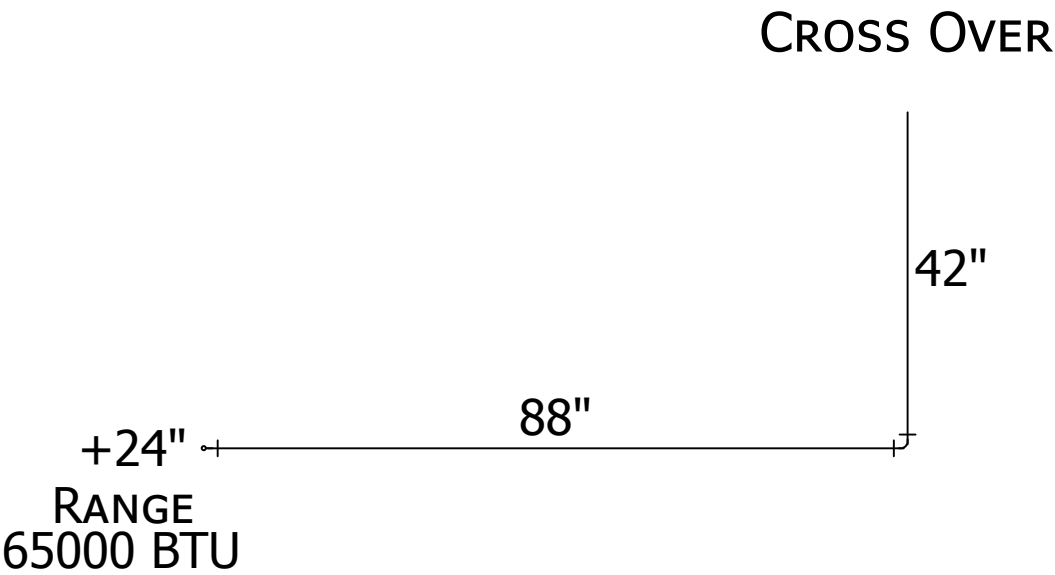
FILENAME: 112APB-2852H32170

SHEET:
G-103

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NOTES:

- 1. ALL PIPE SIZES 1/2" UNLESS OTHERWISE SPECIFIED
- 2. TOTAL BTU = 145,000 (202,000 W/ OPTIONS)
- 3. COLUMN LENGTH = 30'-0"



OPTIONAL SUMMIT KITCHEN

AVAILABLE W/ OPT. STONE COUNTERTOPS

CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

PROJECT:
112APB-2852H32170
52'-0" x 26'-8"
3 BD 2 BT

DRAWN BY: JON RILEY
DATE: 10-21-24
SCALE: N.T.S.

TITLE:
GAS LINE
PLAN

FILENAME: 112APB-2852H32170

SHEET:
G-104

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